

**ZEROSHOT STATE-CONSISTENT IMAGE GENERATION VIA
AGENT FEEDBACK**

by

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Abstract

Zeroshot State-Consistent Image Generation via Agent Feedback

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We study state consistent image generation using agent feedback loop. The focus is to maintain object identity and state across multi-step image sequences (e.g., color, shape, doneness, and presence of tools/containers) by using an agent-style feedback loop that evaluates state consistency and selects improved candidates.

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Chapter 1

Introduction

1.1 Overview

Describe the motivation and problem setting for state-consistent multi-step image generation.

1.2 Approach

Summarize the agent feedback loop, including:

- ground-truth state generation per step (text)
- embedding-based scoring (e.g., CLIP text vs image)
- candidate selection from intermediate generations
- iteration until state consistency improves

1.3 Notes

Add figures/tables by editing the files under `chapters/ch-introduction/tex/`.

Chapter 2

Related Work

2.1 Overview

Describe the motivation and problem setting for state-consistent multi-step image generation.

2.2 Approach

Summarize the agent feedback loop, including:

- ground-truth state generation per step (text)
- embedding-based scoring (e.g., CLIP text vs image)
- candidate selection from intermediate generations
- iteration until state consistency improves

2.3 Notes

Add figures/tables by editing the files under `chapters/ch-lit_review/tex/`.

Chapter 3

Method: Agent Feedback Loop

3.1 Overview

Describe the motivation and problem setting for state-consistent multi-step image generation.

3.2 Approach

Summarize the agent feedback loop, including:

- ground-truth state generation per step (text)
- embedding-based scoring (e.g., CLIP text vs image)
- candidate selection from intermediate generations
- iteration until state consistency improves

3.3 Notes

Add figures/tables by editing the files under `chapters/ch-method/tex/`.

Chapter 4

Experiments

4.1 Overview

Describe the motivation and problem setting for state-consistent multi-step image generation.

4.2 Approach

Summarize the agent feedback loop, including:

- ground-truth state generation per step (text)
- embedding-based scoring (e.g., CLIP text vs image)
- candidate selection from intermediate generations
- iteration until state consistency improves

4.3 Notes

Add figures/tables by editing the files under `chapters/ch-experiments/tex/`.

4.4 CLIP-based State Consistency Evaluation

We report state consistency using CLIP by comparing image embeddings against text embeddings derived from ground-truth state descriptions. In the object-phrase mode, each step is decomposed into five main object/state phrases (e.g., “shimmering

oil”, “intact yolk”). If fewer than five valid phrases are available, the final score is the average cosine similarity across the available phrases.

4.4.1 How To Build A Pc

- Average score: **0.2546**
- Scoring mode: `gt_object_phrases`

CHAPTER 4. EXPERIMENTS

Step	Best candidate (score)	Objects/states (5 phrases)
0	<code>step_0_6.png</code> (0.2583)	budget list on desk; visible intended use note; visible computer monitor; visible PC components list; visible desk surface
1	<code>step_1_5.png</code> (0.2657)	visible labeled CPU; visible compatible motherboard; visible organized RAM; visible noted GPU; visible laid out storage
2	<code>step_2_0.png</code> (0.2408)	visible clean workspace; visible soft mat; visible anti-static wristband; metal screwdriver; covered surface
3	<code>step_3_0.png</code> (0.2719)	visible installed CPU; visible mounted cooler; visible inserted RAM sticks; visible motherboard socket; visible motherboard slots
4	<code>step_4_4.png</code> (0.2738)	visible secured power supply; visible mounted motherboard; inside case; visible screwed motherboard; case with screws
5	<code>step_5_3.png</code> (0.2687)	visible respective slots; visible connected data cables; visible connected power cables; visible placed SSD; visible placed HDD
6	<code>step_6_3.png</code> (0.2728)	visible graphics card inserted; PCIe slot on motherboard; visible additional expansion cards; visible slots; visible motherboard slots
7	<code>step_7_6.png</code> (0.2966)	visible power cables connected; visible front-panel cables connected; visible fan cables connected; visible cables neatly managed; visible cables tied for airflow
8	<code>step_8_3.png</code> (0.2673)	visible powered PC; visible BIOS screen; visible adjusted basic settings; visible component detection; visible powered monitor
9	<code>step_9_2.png</code> (0.2903)	visible installation screen displayed; visible essential software installed; visible stability test running; visible operating system installation; visible monitor dis-

4.4.2 How To Change Car Tyre

- Average score: **0.2267**
- Scoring mode: `gt_object_phrases`

Step	Best candidate (score)	Objects/states (5 phrases)
0	<code>step_0_7.png</code> (0.2476)	visible flat area; visible safe area; visible engaged handbrake; visible flashing hazard lights; visible parked car
1	<code>step_1_0.png</code> (0.2498)	visible heavy objects; visible opposite wheel; visible front wheel; visible behind wheel; visible wheel chocks
2	<code>step_2_8.png</code> (0.2309)	visible slightly loosened wheel nuts; metal wrench; car on the ground; visible grounded car; visible loosened nuts
3	<code>step_3_6.png</code> (0.2274)	visible correct jacking point; visible raised car; visible off-ground tyre; visible positioned jack; visible correct position
4	<code>step_4_0.png</code> (0.2366)	fully removed wheel nuts; visible flat tyre; visible taken off tyre; visible removed nuts; fully removed wheel nuts
5	<code>step_5_4.png</code> (0.2563)	visible spare tyre mounted; visible aligned holes; hub with studs; visible mounted spare tyre; visible aligned studs
6	<code>step_6_2.png</code> (0.2380)	visible hand-tightened nuts; visible criss-cross pattern; visible tightened wheel nuts; visible patterned wheel; visible crossed nuts
7	<code>step_7_8.png</code> (0.2328)	visible carefully lowered car; visible grounded tyre; fully tightened nuts; visible criss-cross pattern; visible lowered car
8	<code>step_8_2.png</code> (0.2365)	visible completely lowered car; visible removed jack; visible removed chocks; visible stowed tools; visible stowed flat tyre

4.4.3 How To Cook A Fried Egg

- Average score: **0.2677**

- Scoring mode: `gt_object_phrases`

Step	Best candidate (score)	Objects/states (5 phrases)
0	<code>step_0_0.png</code> (0.2841)	nonstick pan; visible medium heat stove; empty pan; hot stove; metal pan
1	<code>step_1_1.png</code> (0.2720)	small amount of oil; shimmering oil; foaming butter; small pan; visible added butter
2	<code>step_2_1.png</code> (0.2868)	a small bowl; an intact yolk; a cracked egg; visible no shell pieces; a raw egg
3	<code>step_3_1.png</code> (0.2654)	visible gently slid egg; visible heated pan; intact yolk; visible bowl egg; visible gently slid bowl
4	<code>step_4_1.png</code> (0.2556)	crispy edges; runny yolk; set egg white; slightly crispy edges; beginning to set egg white
5	<code>step_5_1.png</code> (0.2704)	covered pan; cooked egg white; visible yolk; briefly covered pan; top of the egg white
6	<code>step_6_1.png</code> (0.2572)	visible seasoned egg; set whites; visible salted egg; visible peppered egg; fully set whites
7	<code>step_7_0.png</code> (0.2849)	cooked egg; visible carefully lifted spatula; visible ready egg; cooked egg from pan; visible served egg

4.4.4 How To Cook A Pasta

- Average score: **0.2579**
- Scoring mode: `gt_object_phrases`

Step	Best candidate (score)	Objects/states (5 phrases)
0	<code>step_0_0.png</code> (0.2658)	large pot; stove filled; visible pinch of salt; boiling water; visible water begins to boil
1	<code>step_1_0.png</code> (0.2772)	uncooked pasta; boiling water; wooden spoon; visible submerged pasta; visible stirred pasta
2	<code>step_2_0.png</code> (0.2647)	boiling water; visible cooking pasta; visible rising steam; visible stirred pasta; hot water
3	<code>step_3_1.png</code> (0.2344)	visible tender pasta; visible slightly firm texture; visible lifted fork; visible checked doneness; visible piece of pasta
4	<code>step_4_0.png</code> (0.2733)	visible reserved pasta water; visible measuring cup; visible drained pasta; colander in the sink; full cup of water
5	<code>step_5_0.png</code> (0.2468)	visible drained pasta; empty pot; visible added sauce; visible mixed pasta; wooden spoon
6	<code>step_6_1.png</code> (0.2581)	visible served pasta; visible grated cheese; visible fresh herbs; visible topped plate; visible pasta plate
7	(n/a)	visible plated pasta; visible drizzle of olive oil; visible garnished pasta; visible ready pasta; visible eaten pasta

4.4.5 How To Decorate Christmas Tree

- Average score: **0.2393**
- Scoring mode: `gt_object_phrases`

CHAPTER 4. EXPERIMENTS

Step	Best candidate (score)	Objects/states (5 phrases)
0	<code>step_0_8.png</code> (0.2583)	green Christmas tree; visible upright tree; visible living room corner; nearby power outlet; visible stable tree stand
1	<code>step_1_1.png</code> (0.2329)	visible fluffed branches; visible shaped tree; full appearance; visible even branches; visible undecorated tree
2	<code>step_2_9.png</code> (0.2693)	white string lights; visible inner branches; visible outer branches; visible wrapped lights; visible woven lights
3	<code>step_3_4.png</code> (0.2272)	red garland; visible gold garland; visible evenly draped garland; visible consistent spacing; around the tree
4	<code>step_4_6.png</code> (0.2705)	large ornaments; visible round ornaments; red ornaments; visible gold ornaments; visible evenly distributed ornaments
5	<code>step_5_5.png</code> (0.2552)	visible medium ornaments; small ornaments; visible various colors; visible textured tree; visible colorful tree
6	<code>step_6_0.png</code> (0.2305)	visible special ornaments; visible eye level; visible prominent spots; visible family photo; visible tree star
7	<code>step_7_3.png</code> (0.2408)	visible gold star; visible tree topper; visible highest point; visible centered tree; visible stable topper
8	<code>step_8_0.png</code> (0.2444)	red tree skirt; visible base of the tree; visible smooth skirt; visible even skirt; visible tree stand
9	<code>step_9_3.png</code> (0.2618)	fully decorated tree; visible adjusted decorations; visible symmetrical tree; visible balanced tree; visible festive tree

4.4.6 How To Do Cpr

- Average score: **0.2416**
- Scoring mode: **gt_object_phrases**

Step	Best candidate (score)	Objects/states (5 phrases)
0	step_0_9.png (0.2240)	visible unresponsive person; visible firm surface; visible flat surface; visible clear scene; visible hazard-free area
1	step_1_6.png (0.2588)	visible used phone; visible emergency services call; visible retrieved AED; nearby location; visible emergency phone
2	step_2_6.png (0.2190)	visible exposed chest; visible flat surface; visible person's back; remains on back; visible chest is exposed
3	step_3_7.png (0.2707)	visible centered hands; visible consistent rhythm; visible person's chest; visible chest compressions; visible positioned hands
4	step_4_3.png (0.2376)	visible tilted head; visible chin-lift position; visible rescue breaths; visible chest rise; visible person's head
5	step_5_9.png (0.2716)	visible chest compressions; visible rescue breaths; visible uninterrupted cycles; visible the person's back; visible 30 compressions
6	step_6_6.png (0.2540)	turned on AED; visible attached pads; visible person's chest; visible delivered shock; visible immediate chest compressions
7	step_7_3.png (0.2627)	visible urgent CPR; visible lifeless person; visible emergency help; visible continuous AED use; visible desperate situation

4.4.7 How To Fold A Paper Crane

- Average score: **0.2536**
- Scoring mode: **gt_object_phrases**

Step	Best candidate (score)	Objects/states (5 phrases)
0	step_0_8.png (0.2831)	visible square sheet; visible colored side; visible X crease; visible flat paper; visible diagonal folds
1	step_1_2.png (0.2806)	visible flipped paper; visible folded paper; visible unfolded paper; visible square base; visible diagonal creases
2	step_2_6.png (0.2540)	open end; visible lower edges; visible center crease; visible top triangle; visible bottom corner
3	step_3_3.png (0.2754)	visible diamond's front flap; visible lower edges folded; visible long kite shape; visible center folded edges; visible back repeated shape
4	step_4_0.png (0.2486)	visible long bottom points; visible existing creases; visible well-defined creases; visible folded upward points; visible head and tail
5	step_5_3.png (0.2603)	visible reverse folded point; visible crane's head; visible kite shape paper; visible folded down tip; inside point
6	step_6_7.png (0.2663)	visible gently pulled wings; visible horizontal position wings; visible upright crane's body; visible adjusted crane's body; visible horizontal wings
7	step_7_0.png (0.2386)	visible complete paper crane; visible spread wings; visible standing base; visible distinct head; visible distinct tail

4.4.8 How To Install Curtain

- Average score: **0.2604**
- Scoring mode: **gt_object_phrases**

Step	Best candidate (score)	Objects/states (5 phrases)
0	step_0_1.png (0.2600)	visible extended tape measure; visible window frame; nearby notepad; visible written measurements; visible across window
1	step_1_0.png (0.2249)	visible pencil marks; visible level against the wall; visible even marks; pencil in hand; marks above the window
2	step_2_0.png (0.2412)	visible marked spots; small holes; visible wall anchors; visible positioned drill; visible drilled wall
3	step_3_1.png (0.2579)	metal brackets; visible attached brackets; visible secure brackets; a screwdriver in use; visible level brackets
4	step_4_0.png (0.3049)	visible threaded curtain rod; visible attached curtain rings; visible gathered curtain; metal rod; visible end of curtain
5	step_5_0.png (0.2843)	metal curtain rod; secured set screws; visible hanging curtain; visible tightened screws; visible mounted brackets
6	step_6_0.png (0.2889)	visible evenly hanging curtain panels; fully covered window; visible smooth curtain fabric; visible straight curtain panels; visible adjusted curtain panels

4.4.9 How To Make Cheesy Garlic Pull-Apart Bread

- Average score: **0.2622**
- Scoring mode: **gt_object_phrases**

CHAPTER 4. EXPERIMENTS

Step	Best candidate (score)	Objects/states (5 phrases)
0	<code>step_0_14.png</code> (0.2475)	visible a loaf of bread; visible a stick of butter; visible garlic cloves; visible shredded cheese; a small bowl
1	<code>step_1_14.png</code> (0.2941)	visible preheating oven; visible lined baking dish; visible parchment paper; baking dish on counter; hot oven
2	<code>step_2_11.png</code> (0.2664)	melted butter; visible minced garlic; visible mixed seasonings; visible fresh parsley; visible seasoned salt
3	<code>step_3_2.png</code> (0.2526)	visible loaf of bread; visible crosshatch pattern; visible deep cuts; visible bottom crust; visible sliced loaf
4	<code>step_4_7.png</code> (0.2909)	visible gently pulled bread; visible garlic butter; visible spooned cuts; visible bread sections; gaps in bread
5	<code>step_5_6.png</code> (0.2696)	visible shredded cheese; visible evenly stuffed cheese; visible bread pieces; visible slits between bread; visible stuffed slits
6	<code>step_6_16.png</code> (0.2977)	loaf in baking dish; loosely covered foil; baking dish in oven; visible placed baking dish; covered baking dish
7	<code>step_7_2.png</code> (0.2588)	visible removed foil; golden top; visible slightly crisp bread; visible baked loaf; visible baked bread
8	<code>step_8_6.png</code> (0.2940)	visible cooling bread; visible extra parsley; visible oven bread; visible sprinkled parsley; visible briefly cooling bread
9	<code>step_9_0.png</code> (0.2546)	visible cheesy bread; visible garlic bread; warm bread; visible easily pulled sections; visible apart sections

4.4.10 How To Raise A Child

- Average score: **0.2286**
- Scoring mode: `gt_object_phrases`

CHAPTER 4. EXPERIMENTS

Step	Best candidate (score)	Objects/states (5 phrases)
0	<code>step_0_7.png</code> (0.2422)	visible cozy living room; visible sitting child; visible hugging parent; visible child's lap; visible parent's embrace
1	<code>step_1_9.png</code> (0.2054)	nightlight on; visible colorful book; visible bedtime story; child in bed; visible parent reading
2	<code>step_2_2.png</code> (0.2715)	small toothbrush; visible brightly lit bathroom; visible supervising parent; visible child brushing teeth; teeth with toothbrush
3	<code>step_3_5.png</code> (0.2396)	a chart with rules; consequences on the wall; visible a child pointing; visible a parent explaining; a rule on the wall
4	<code>step_4_4.png</code> (0.2453)	visible sitting child; table with books; magnifying glass; visible picture book; visible about animals
5	<code>step_5_3.png</code> (0.2363)	visible smiling parent; visible polite conversation; visible watching child; visible greeting neighbor; visible modeled behavior
6	<code>step_6_2.png</code> (0.2340)	visible other children; visible a sandy sandbox; visible shared toys; a large sandcastle; visible playing child
7	<code>step_7_9.png</code> (0.2482)	visible colorful plate; visible healthy smoothie; visible fruits and vegetables; visible kitchen parent; visible eating child
8	<code>step_8_3.png</code> (0.1912)	small watering can; visible watering plants; visible garden plants; next plants; visible choosing plants
9	<code>step_9_8.png</code> (0.2338)	drawing on a chalkboard; visible kneeling parent; visible listening parent; visible discussing drawing; visible child drawing

Chapter 5

Conclusion

5.1 Overview

Describe the motivation and problem setting for state-consistent multi-step image generation.

5.2 Approach

Summarize the agent feedback loop, including:

- ground-truth state generation per step (text)
- embedding-based scoring (e.g., CLIP text vs image)
- candidate selection from intermediate generations
- iteration until state consistency improves

5.3 Notes

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